

Prasad Namdev Pawar

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Education

Indian Institute of Technology Gandhinagar

Ph.D. Physics

Quantum Optics Group

Gandhinagar, India

Dec 2022 – Present

Indian Institute of Technology Palakkad

M.Sc. Physics

Palakkad, India

Oct 2020 – May 2022

D.G. Ruparel College

B.Sc. Physics

Mumbai, India

Jul 2017 – Oct 2020

Research

Publication

Pawar, P., Bhattacharyya, A., & Venkatesh, B. P. (2025). Comparative Study of Indicators of Chaos in the Closed and Open Dicke Model. *Phys. Rev. E* **113**, 044205. doi.org/10.1103/786l-8zx4

Research Experience

Working in the fields of quantum many-body systems, including quantum chaos, measurement-induced phase transitions, random quantum circuits, and ultracold atoms.

Conferences

- Quantum Trajectories 2025 (Jan 2025, ICTS Bengaluru)
www.icts.res.in/program/qt
- Quantum Technologies using Ultra-Cold Atoms (QTUCA) 2024 (Sep 2024, IISER Pune)
sites.google.com/view/qtdua/home
- Intl. Conf. on Quantum Technologies and Applications (ICQTA) 2024 (Feb 2024, MAHE Manipal)
conference.manipal.edu/ICQTA2024
- Intl. Conf. on Photonics Quantum Information and Quantum Communication (Jan 2024, SNBNCBS Kolkata)
www.bose.res.in/Conferences/PQIQC23
- Introduction to Precision Measurements and Quantum Metrology (Jul 2023, ICTS Bengaluru)
www.icts.res.in/program/PMQM-2023

Presented posters on quantum chaos at ICPQIQ-2024, ICQTA-2024, QTUCA-2024, and Quantum Trajectories 2025.

Honors / Awards

Best Poster Award (Physics) in PhD Research Showcase (Aug 2025) at IIT Gandhinagar for quantum chaos in a light-matter system.

Ongoing Projects

Measurement-Induced Phase Transition (MIPT) in Collective Systems

MIPT is the transition from volume-law to area-law entanglement in random quantum circuits controlled by the measurement rate. These models map to classical statistical mechanics models with distinct universality classes. We investigate whether collective measurements alter the universality class of the phase transition. Understanding these informational phases of matter has implications for quantum technology platforms.

Bistability and Non-Equilibrium Steady States (NESS) in Dissipative Bosonic Lattices

The phenomena of bistability and NESS in lattice systems with ultracold atoms remain poorly understood across the Mott–Superfluid regimes. We employ Gutzwiller approximation and SU(3) truncated Wigner approximation to develop a theoretical description. This work aids in understanding ultracold quantum-matter platforms and informing quantum technologies based on them.

Teaching Experience

IIT Gandhinagar

- Graduate Teaching Fellow (GTF) for undergraduate writing course HS 191 (excellent student feedback).
- Teaching assistant for Quantum Mechanics (PH 202), Statistical Mechanics (PH 507), and Quantum Optics (PH 616).

IIT Palakkad

- Teaching assistant for Bachelor's-level Electromagnetism.

Workshops & Examinations

Quantum Computing Workshop & Hackathon

The Eigensolvers and ISTE, NIT Calicut | Intermediate Tier Certificate

Jul 2021

Joint Admission Test for Masters (JAM 2020)

Indian Institute of Technology Kanpur | All India Rank: 572 (Total Candidates: 16 379)

National Graduate Physics Examination (NGPE 2020)

Indian Association of Physics Teachers | Certificate of Merit – National Top 1% (Total Candidates: 11 638)

Other Activities

Nyasa, IIT Gandhinagar

Volunteer for Social Service | initiatives.iitgn.ac.in/nyasa

May 2024 – Present

Quantum Computing Club, IIT Palakkad

Founding Club Secretary

Sep 2021 – Mar 2022

Nehru Science Centre, Mumbai

Internship in Science Communication

Oct 2019 – Dec 2019

VIBGYOR – Ruparel College Physics Fest

Organizing Team Member for Science Communication

Feb 2018 & 2019

Technical Skills

L^AT_EX, Python, Monte Carlo Simulation, Qiskit, Jupyter Notebook, C, Matlab

Languages

Marathi (Native), Hindi (Native), English (Full Professional Proficiency)